

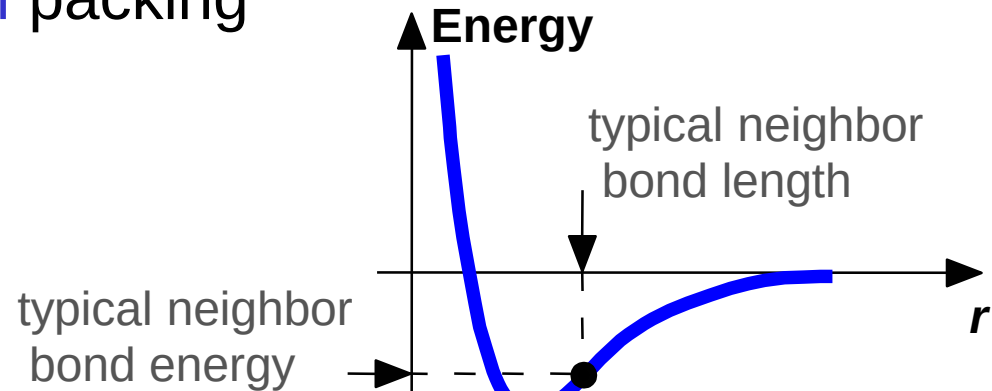
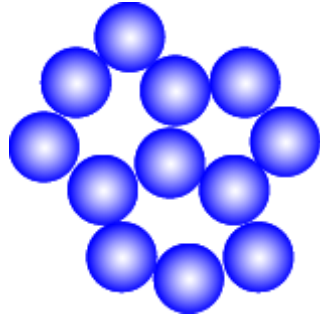
Chapter 3: The Structure of Crystalline Solids

ISSUES TO ADDRESS...

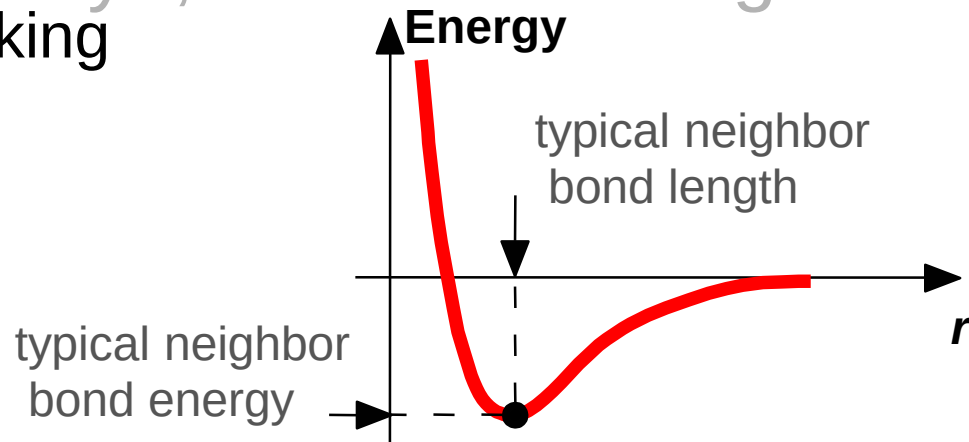
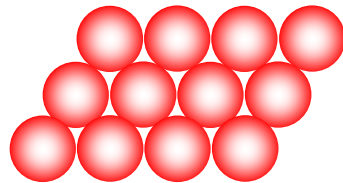
- How do atoms assemble into solid structures?
- How does the density of a material depend on its structure?
- When do material properties vary with the sample (i.e., part) orientation?

Energy and Packing

- Non dense, **random** packing



- Dense, **ordered** packing

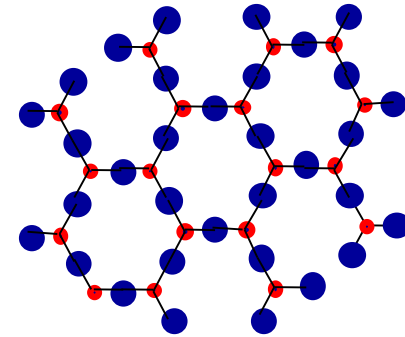


Dense, ordered packed structures tend to have lower energies.

Materials and Packing

Crystalline materials...

- atoms pack in periodic, 3D arrays
- typical of:
 - metals
 - many ceramics
 - some polymers



crystalline SiO₂

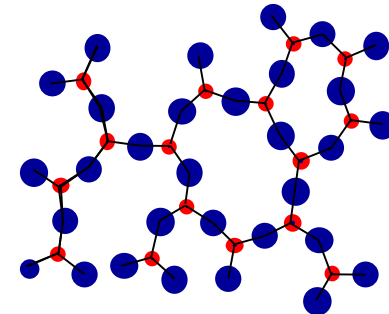
Adapted from Fig. 3.23(a),
Callister & Rethwisch 3e.

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• **Si** • **Oxygen**

Noncrystalline materials...

- atoms have no periodic packing
- occurs for:
 - complex structures
 - rapid cooling



noncrystalline SiO₂

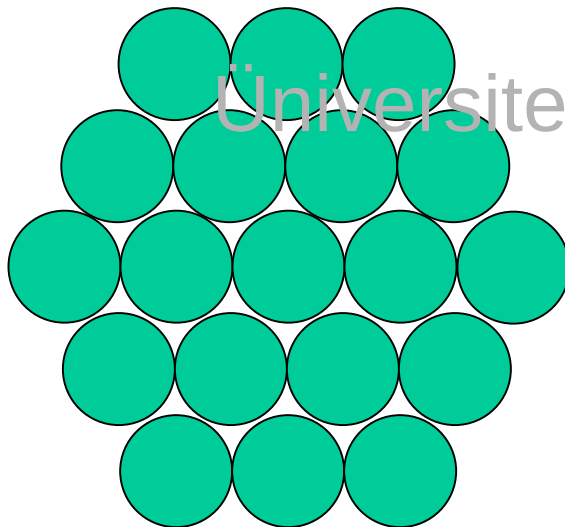
Adapted from Fig. 3.23(b),
Callister & Rethwisch 8e.

"Amorphous" = Noncrystalline

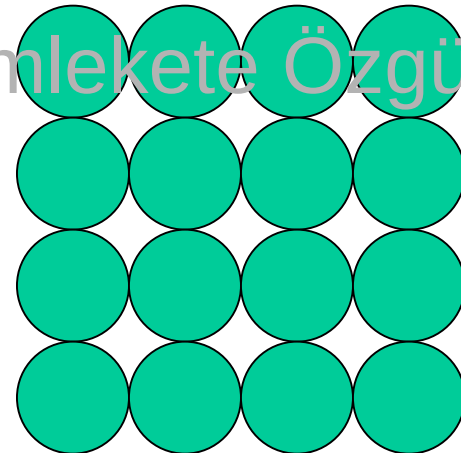
Metallic Crystal Structures

- How can we stack metal atoms to minimize empty space?

2-dimensions



vs.



Now stack these 2-D layers to make 3-D structures

Metallic Crystal Structures

- Tend to be densely packed.
- Reasons for dense packing:
 - Typically, only one element is present, so all atomic radii are the same.
 - Metallic bonding is not directional.
 - Nearest neighbor distances tend to be small in order to lower bond energy.
 - Electron cloud shields cores from each other
- Have the simplest crystal structures.

We will examine three such structures...

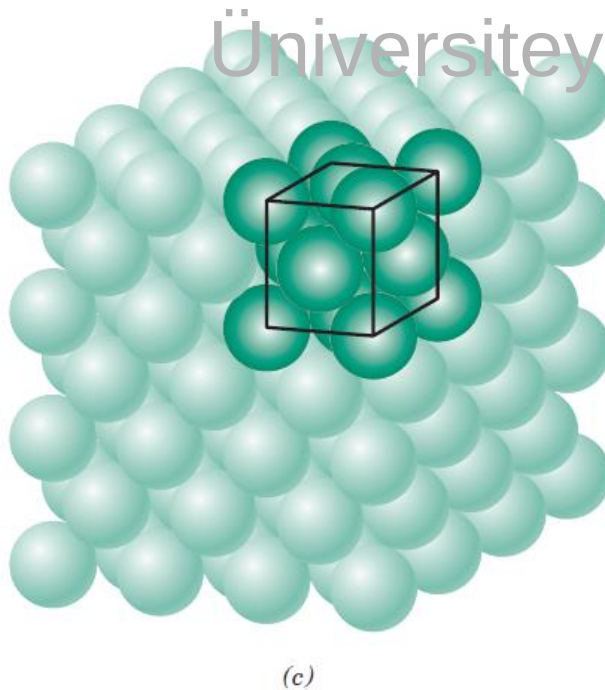
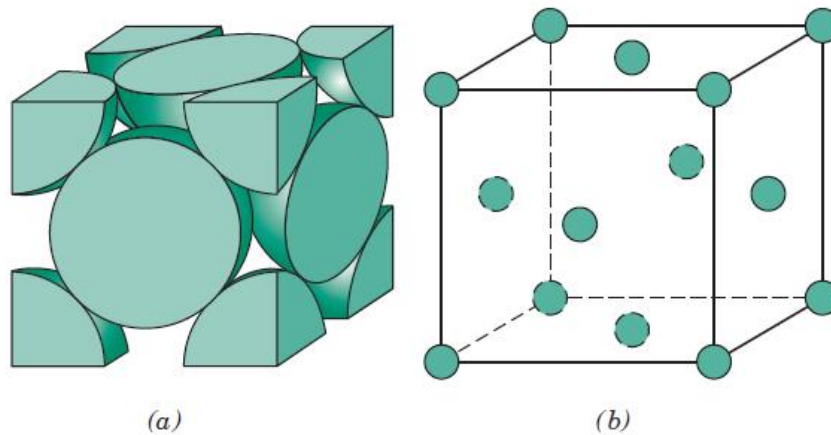
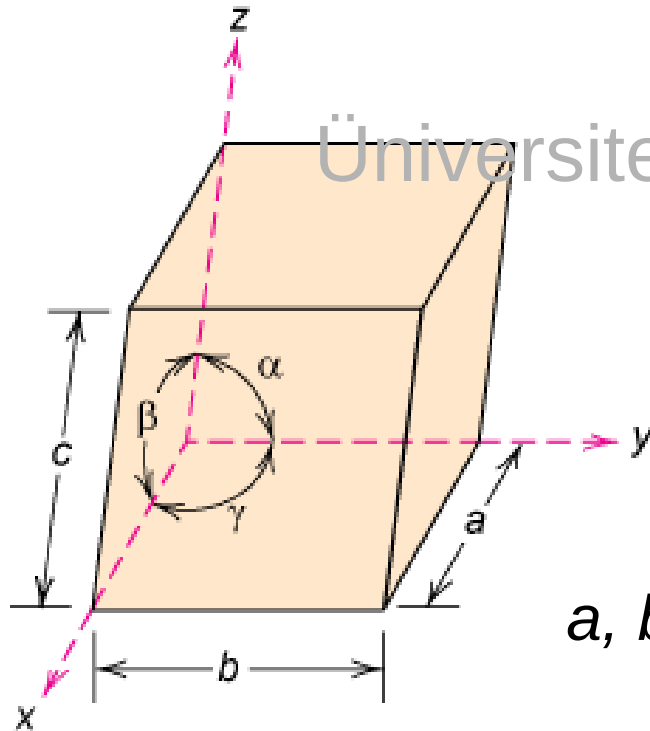


Figure 3.1 For the face-centered cubic crystal structure, (a) a hard-sphere unit cell representation, (b) a reduced-sphere unit cell, and (c) an aggregate of many atoms. [Figure (c) adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulfi, *The Structure and Properties of Materials*, Vol. I, *Structure*, p. 51. Copyright © 1964 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.]

Crystal Systems

Unit cell: smallest repetitive volume which contains the complete lattice pattern of a crystal.



7 crystal systems

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14 crystal lattices

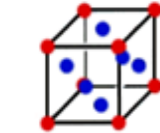
a , b , and c are the lattice constants

Fig. 3.4, Callister & Rethwisch 8e.

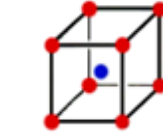
The 14 Bravais Lattices



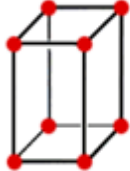
Simple cubic



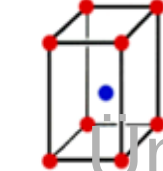
Face-centered cubic



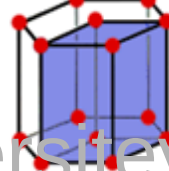
Body-centered cubic



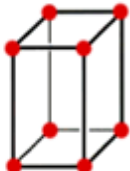
Simple tetragonal



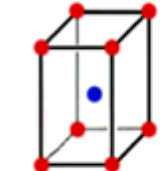
Body-centered tetragonal



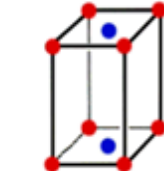
Hexagonal



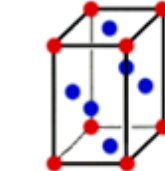
Simple orthorhombic



Body-centered orthorhombic



Base-centered orthorhombic



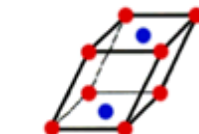
Face-centered orthorhombic



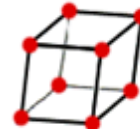
Rhombohedral



Simple Monoclinic



Base-centered monoclinic



Triclinic

Crystal class

Axis system

Cubic

$$a = b = c, \alpha = \beta = \gamma = 90^\circ$$

Tetragonal

$$a = b \neq c, \alpha = \beta = \gamma = 90^\circ$$

Hexagonal

$$a = b \neq c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$$

Rhombohedral

$$a = b = c, \alpha = \beta = \gamma \neq 90^\circ$$

Orthorhombic

$$a \neq b \neq c, \alpha = \beta = \gamma = 90^\circ$$

Monoclinic

$$a \neq b \neq c, \alpha = \gamma = 90^\circ, \beta \neq 90^\circ$$

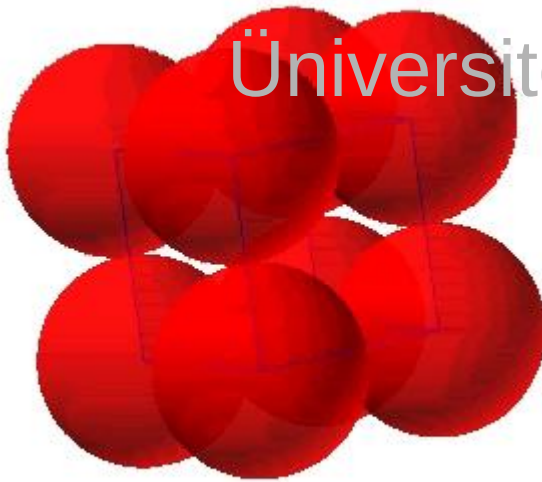
Triclinic

$$a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^\circ$$

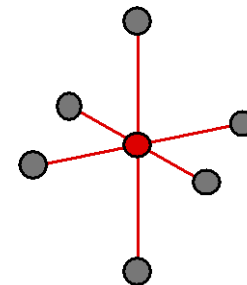
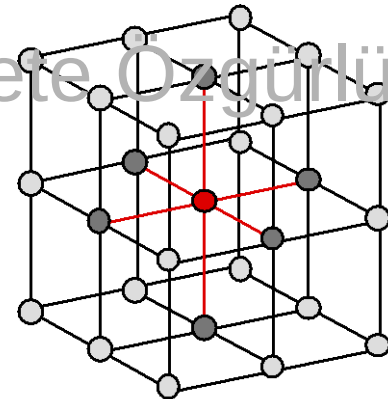
Simple Cubic Structure (SC)

- Rare due to low packing density (only Po has this structure)
- **Close-packed directions** are cube edges.

- **Coordination #** = 6
(# nearest neighbors)



Click once on image to start animation
(Courtesy P.M. Anderson)

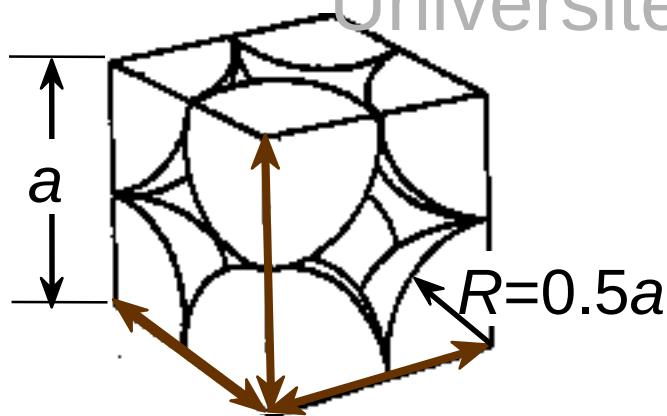


Atomic Packing Factor (APF)

$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

contains $8 \times 1/8 =$

1 atom/unit cell

Adapted from Fig. 3.24,
Callister & Rethwisch 8e.

$$APF = \frac{\overbrace{1}^{\text{atoms}} \overbrace{\frac{4}{3} \pi (0.5a)^3}^{\text{volume atom}}}{\underbrace{a^3}_{\text{volume unit cell}}}$$

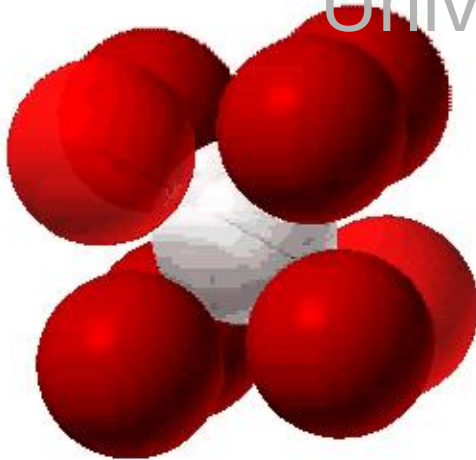
Body Centered Cubic Structure (BCC)

- Atoms touch each other along cube diagonals.
--Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.

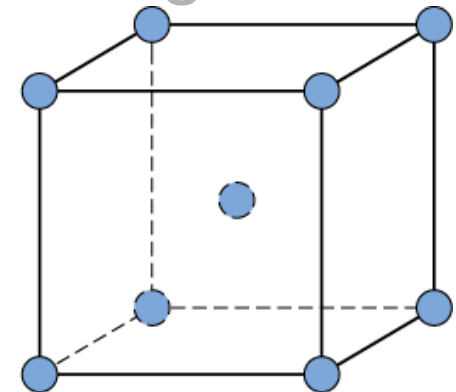
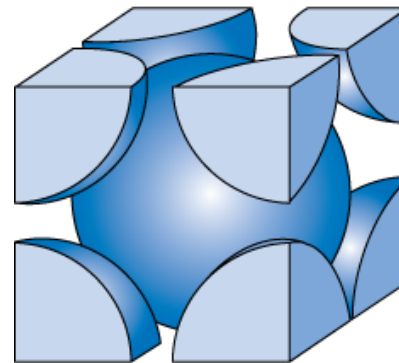
ex: Cr, W, Fe (α), Tantalum, Molybdenum

- Coordination # = 8

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Click once on image to start animation
(Courtesy P.M. Anderson)

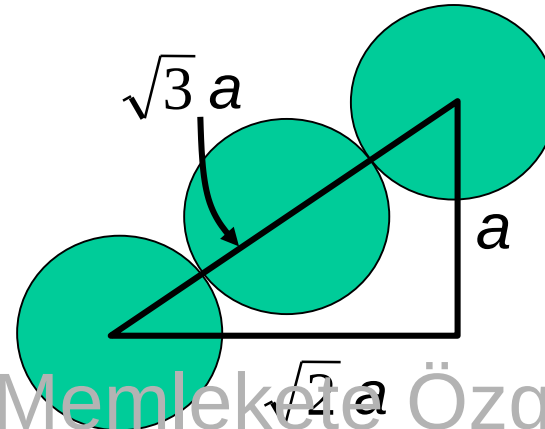
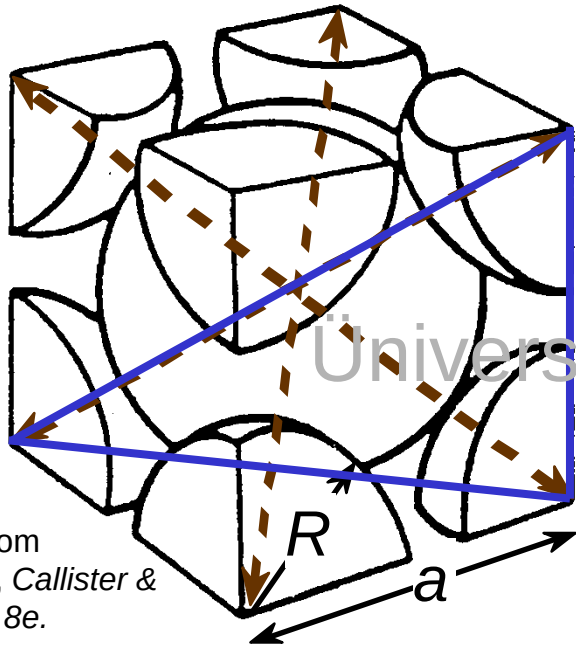


Adapted from Fig. 3.2,
Callister & Rethwisch 8e.

2 atoms/unit cell: 1 center + 8 corners $\times$ 1/8

Atomic Packing Factor: BCC

- APF for a body-centered cubic structure = 0.68



Close-packed directions:
length = $4R = \sqrt{3} a$

$$\text{APF} = \frac{\begin{array}{c} \text{atoms} \\ \text{unit cell} \end{array} \cdot \begin{array}{c} \text{volume} \\ \text{atom} \end{array}}{\begin{array}{c} \text{volume} \\ \text{unit cell} \end{array}}$$

$$\text{APF} = \frac{2 \cdot \frac{4}{3} \pi (\sqrt{3}a/4)^3}{a^3}$$

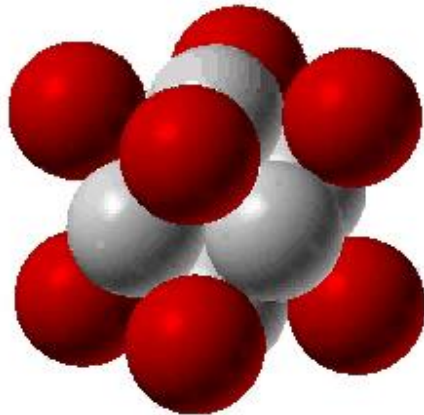
Face Centered Cubic Structure (FCC)

- Atoms touch each other along face diagonals.
--Note: All atoms are identical; the face-centered atoms are shaded differently only for ease of viewing.

ex: Al, Cu, Au, Pb, Ni, Pt, Ag

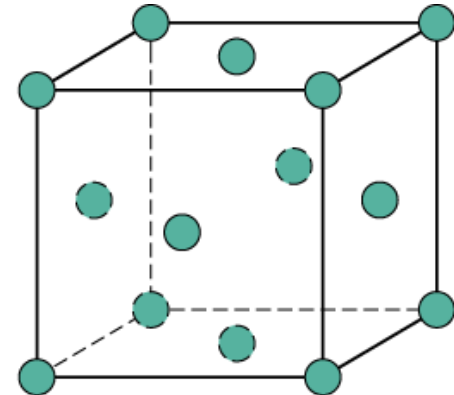
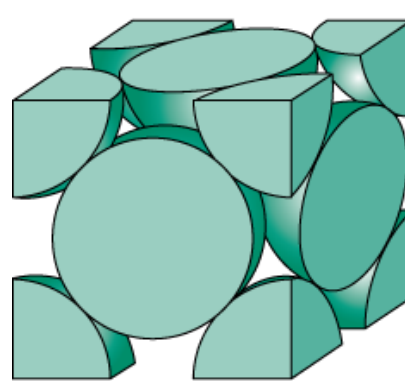
- Coordination # = 12

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Click once on image to start animation

(Courtesy P.M. Anderson)

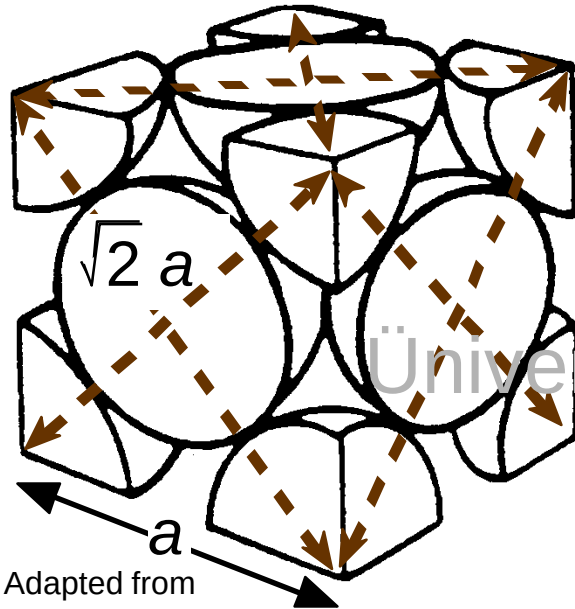


Adapted from Fig. 3.1, Callister & Rethwisch 8e.

4 atoms/unit cell: $6 \text{ face} \times \frac{1}{2} + 8 \text{ corners} \times \frac{1}{8}$

Atomic Packing Factor: FCC

- APF for a face-centered cubic structure = 0.74
maximum achievable APF



Adapted from
Fig. 3.1(a),
Callister &
Rethwisch 8e.

Close-packed directions:
length = $4R = \sqrt{2} a$

Unit cell contains:
 $6 \times 1/2 + 8 \times 1/8$
= **4 atoms/unit cell**

$$\text{APF} = \frac{\begin{array}{c} \text{atoms} \\ \text{unit cell} \end{array} \quad 4 \quad \frac{4}{3} \pi (\sqrt{2}a/4)^3 \quad \begin{array}{c} \text{volume} \\ \text{atom} \end{array}}{\begin{array}{c} \text{volume} \\ \text{unit cell} \end{array} \quad a^3}$$

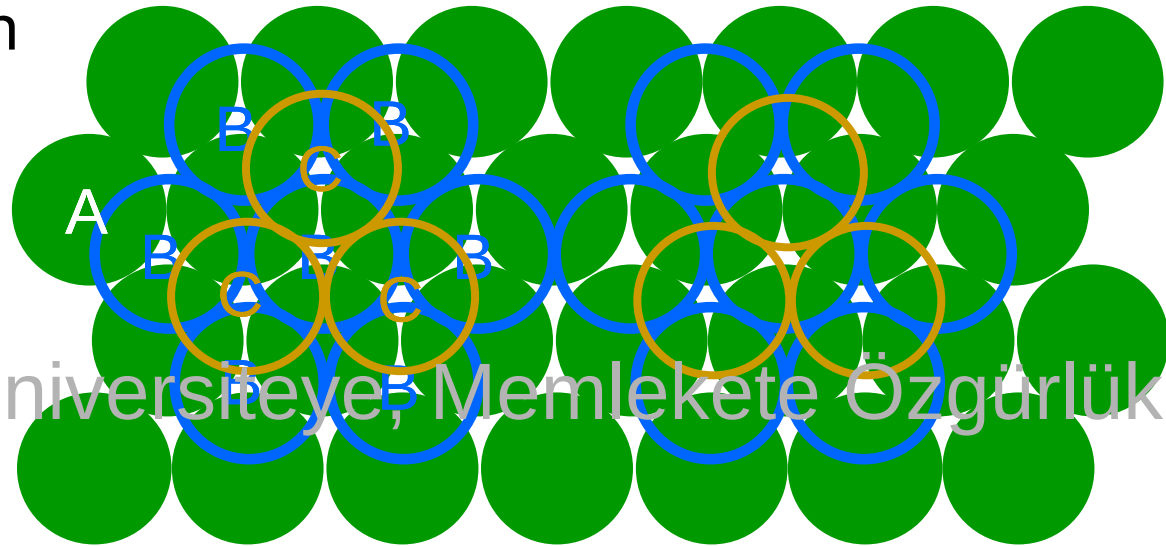
FCC Stacking Sequence

- ABCABC... Stacking Sequence
- 2D Projection

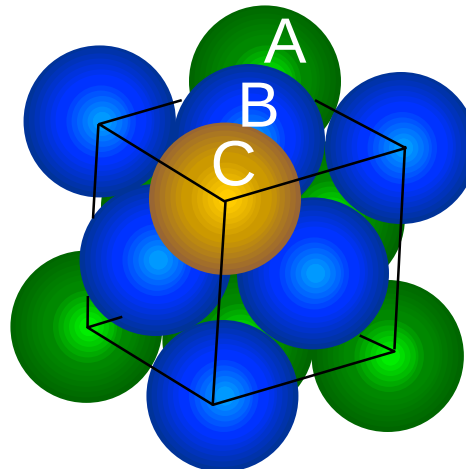
A sites

B sites

C sites



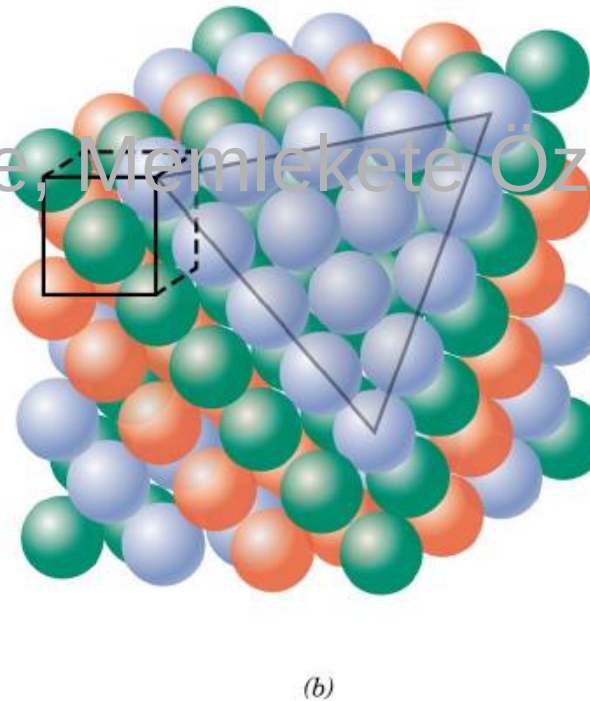
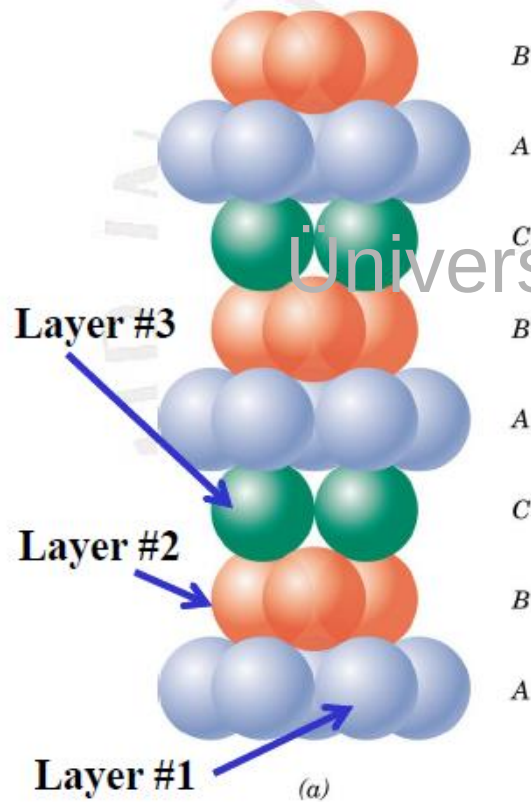
- FCC Unit Cell



Face Centered Cubic Packed Structure (FCC)

•Stacking sequence is ABCABCABCABCABCABC.....

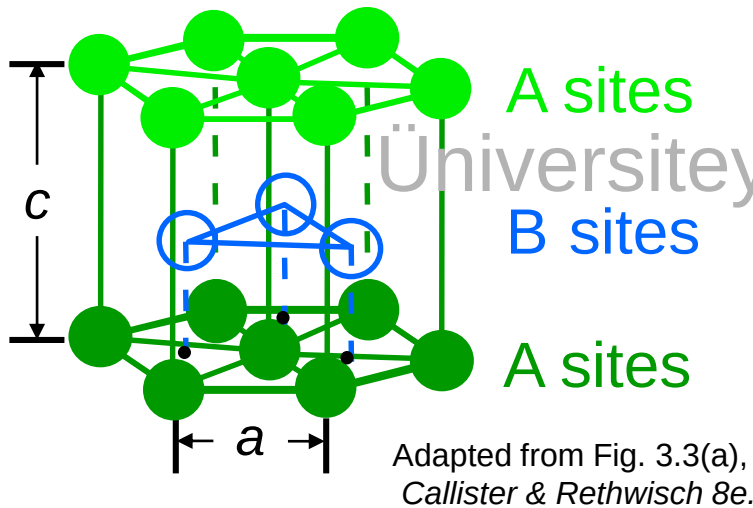
•APF: 74%



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Hexagonal Close-Packed Structure (HCP)

- ABAB... Stacking Sequence
- 3D Projection



- 2D Projection



- Coordination # = 12

- APF = 0.74

- $c/a = 1.633$

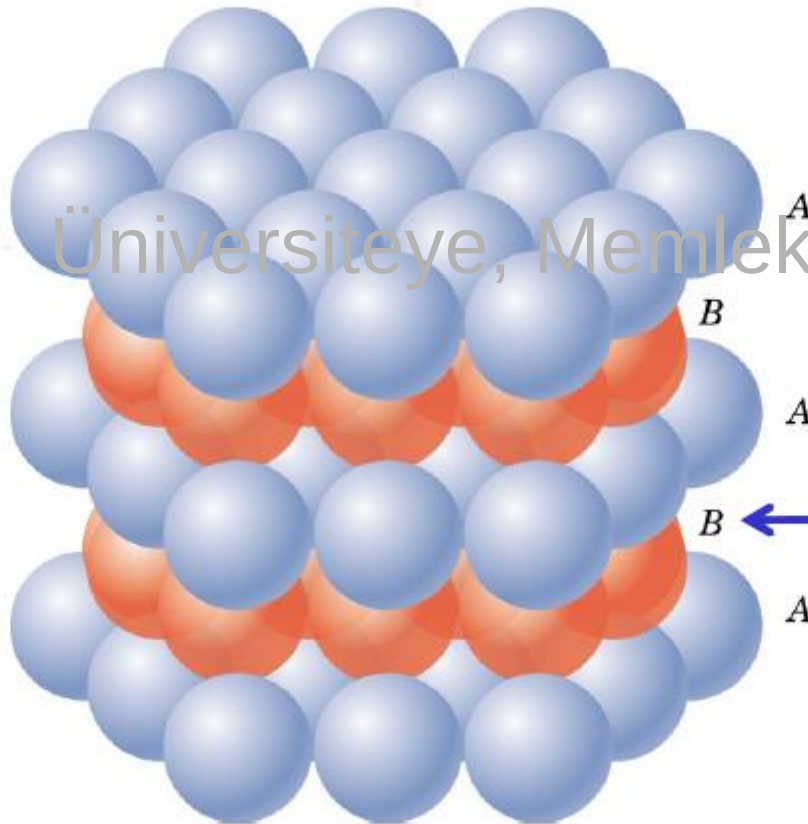
6 atoms/unit cell

ex: Cd, Mg, Ti, Zn

Hexagonal Close Packed Structure (HCP)

•Stacking sequence is ABABABABABABA.....

Ex:
Cadmium,
Titanium...



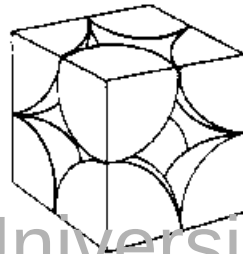
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•APF: 74%

Summary of packing types for metallic structures

Type of Packing	Packing Efficiency	Coordination Number
-----------------	--------------------	---------------------

Simple cubic (sc)

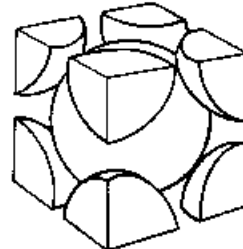


52%

6

$$a = 2R$$

Body-centered cubic (bcc)



68%

8

$$a = \frac{4R}{\sqrt{3}}$$

Hexagonal close-packed (hcp)

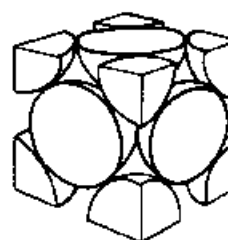
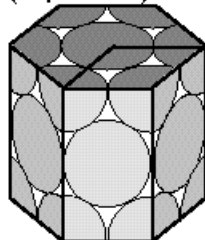
74%

12

Cubic close-packed (ccp or fcc)

74%

12



$$a = 2R\sqrt{2}$$

Theoretical Density, ρ

$$\text{Density} = \rho = \frac{\text{Mass of Atoms in Unit Cell}}{\text{Total Volume of Unit Cell}}$$

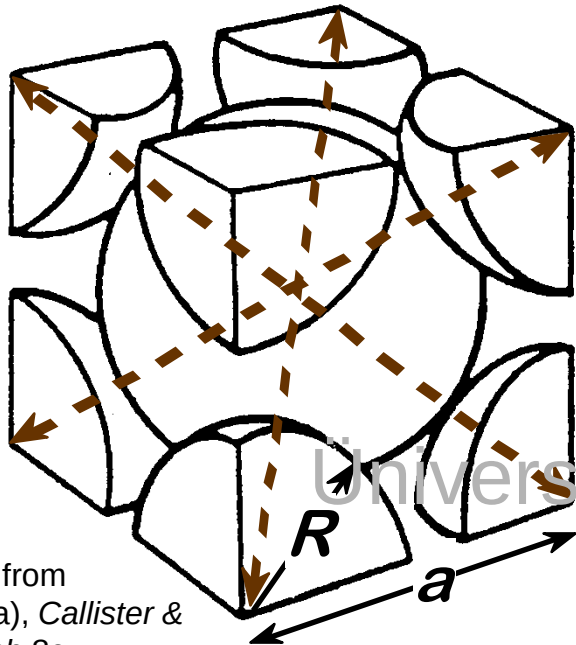
$$\rho = \frac{n A}{V_c N_A}$$

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where

- n = number of atoms/unit cell
- A = atomic weight
- V_c = Volume of unit cell = a^3 for cubic
- N_A = Avogadro's number
- = 6.022×10^{23} atoms/mol

Theoretical Density, ρ



Adapted from
Fig. 3.2(a), Callister &
Rethwisch 8e.

- Ex: Cr (BCC)

$$A = 52.00 \text{ g/mol}$$

$$R = 0.125 \text{ nm}$$

$$n = 2 \text{ atoms/unit cell}$$

$$a = 4R/\sqrt{3} = 0.2887 \text{ nm}$$

$$\rho = \frac{\frac{\text{atoms}}{\text{unit cell}} \cdot A}{\frac{\text{volume}}{\text{unit cell}} \cdot \frac{\text{atoms}}{\text{mol}}}$$

$\frac{\text{atoms}}{\text{unit cell}} = 2$
 $A = 52.00 \text{ g/mol}$
 $\frac{\text{volume}}{\text{unit cell}} = a^3$
 $\frac{\text{atoms}}{\text{mol}} = 6.022 \times 10^{23}$

$$\rho_{\text{theoretical}} = 7.18 \text{ g/cm}^3$$

$$\rho_{\text{actual}} = 7.19 \text{ g/cm}^3$$

- Ex.

Calculate the radius of an iridium atom, given that Ir has an FCC crystal structure, a density of 22.4 g/cm^3 , and an atomic weight of 192.2 g/mol .

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Densities of Material Classes

In general

$$\rho_{\text{metals}} > \rho_{\text{ceramics}} > \rho_{\text{polymers}}$$

Why?

Metals have...

- close-packing (metallic bonding)
- often large atomic masses

Ceramics have...

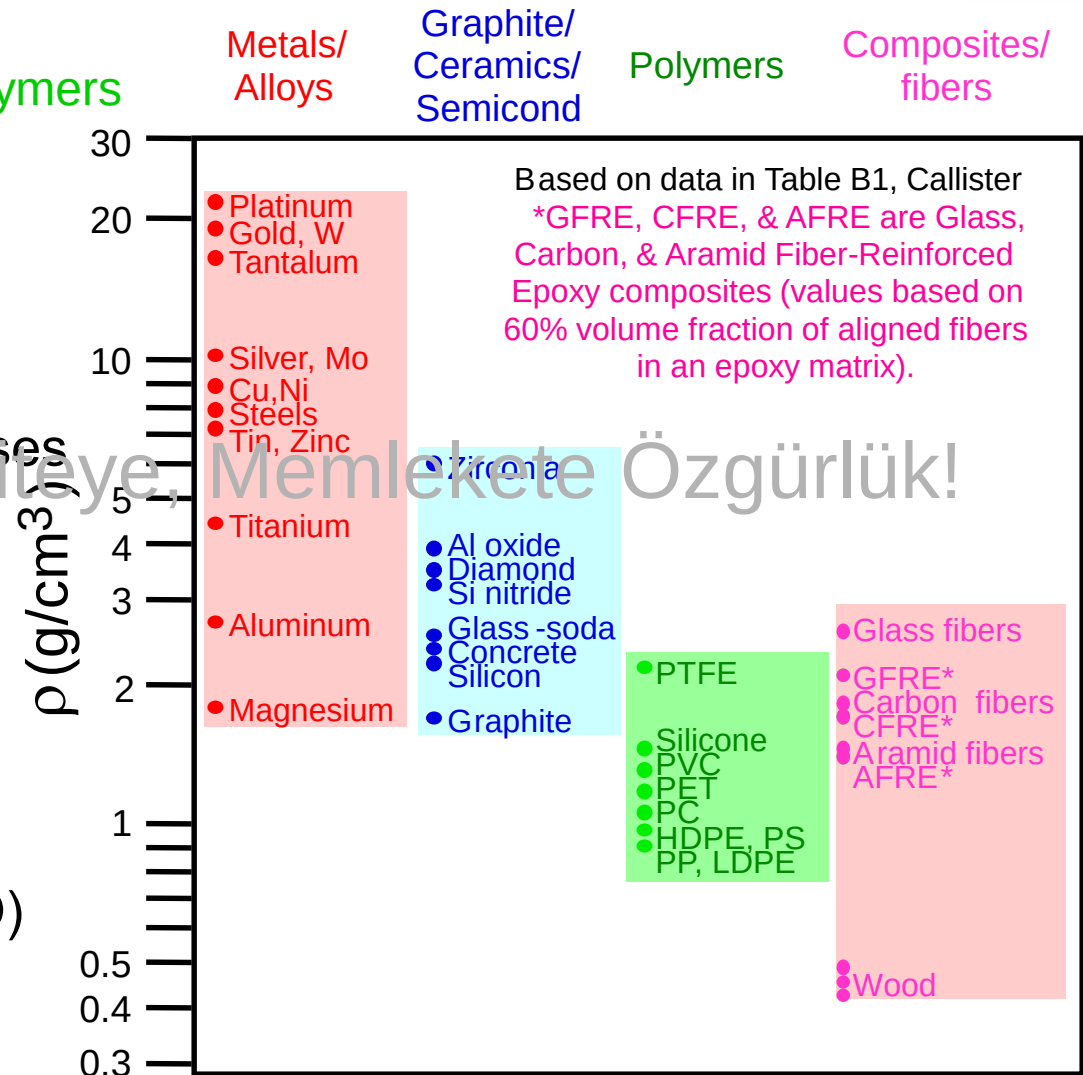
- less dense packing
- often lighter elements

Polymers have...

- low packing density (often amorphous)
- lighter elements (C,H,O)

Composites have...

- intermediate values

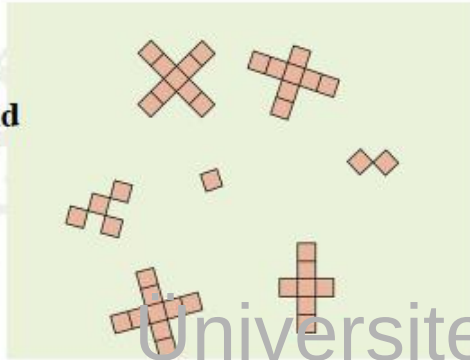


Data from Table B.1, Callister & Rethwisch, 8e.

Crystals as Building Blocks

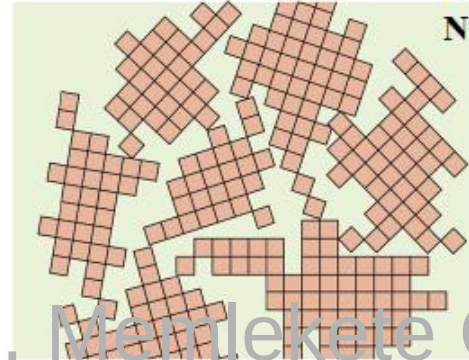
“Crystallization: Solidification”

Small crystal
Nuclei in liquid



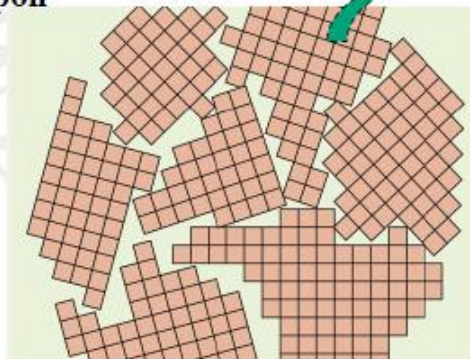
(a)

Growth of
Nuclei in liquid



(b)

Formation of many
“**GRAINS**” upon
Solidification



(c)

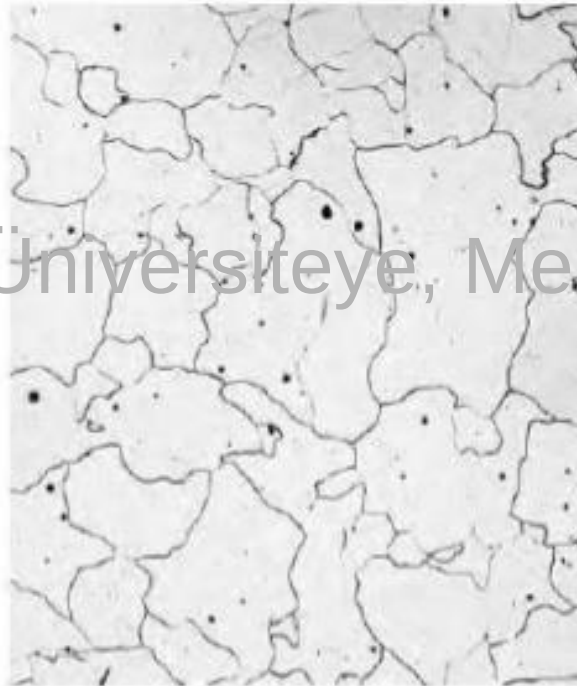
Polycrystalline
microstructure



(d)

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Figure 9.25
Photomicrographs of
(a) α ferrite (90 $\times$)
and (b) austenite
(325 $\times$). (Copyright
1971 by United
States Steel
Corporation.)



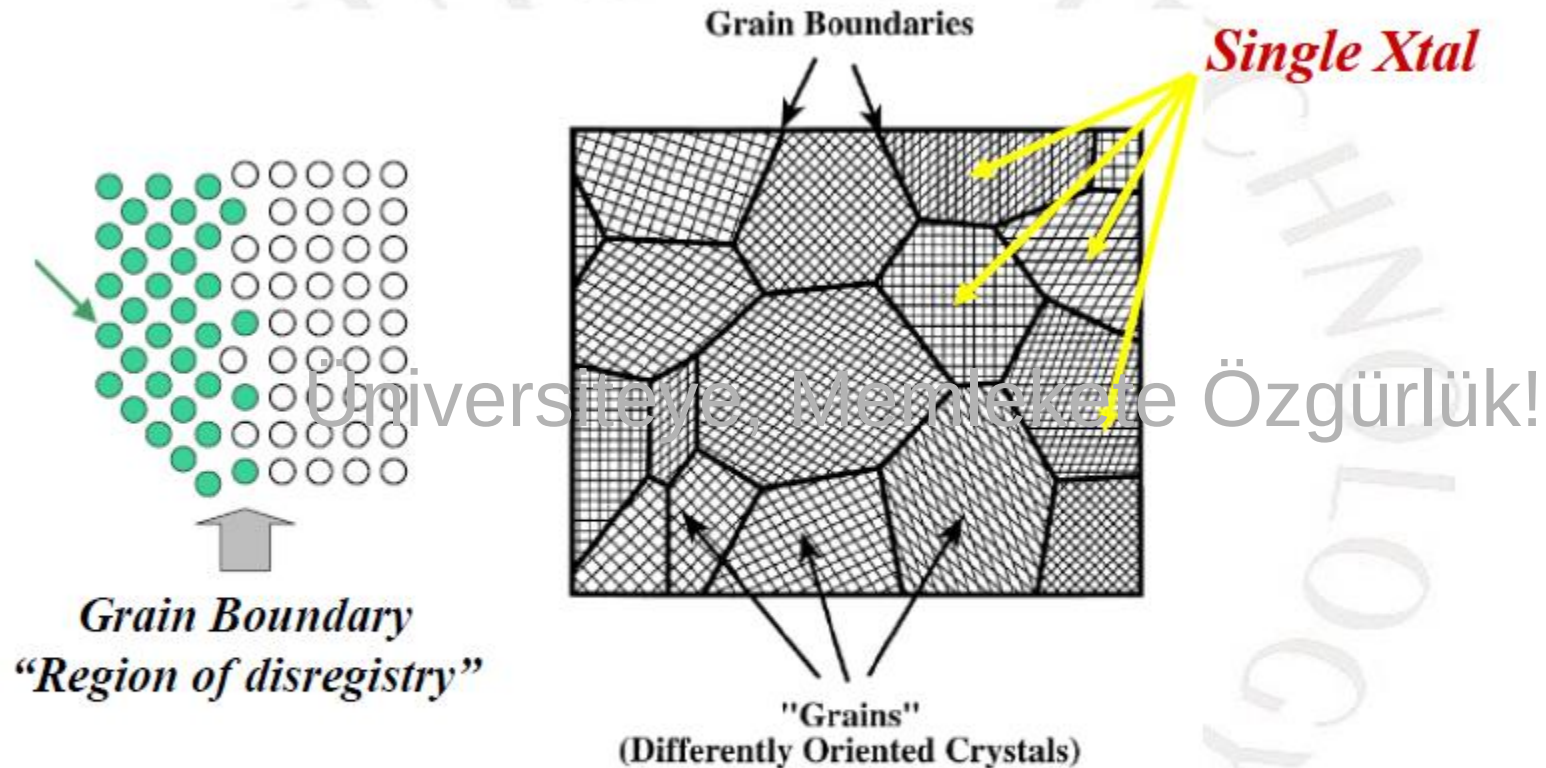
(a)



(b)

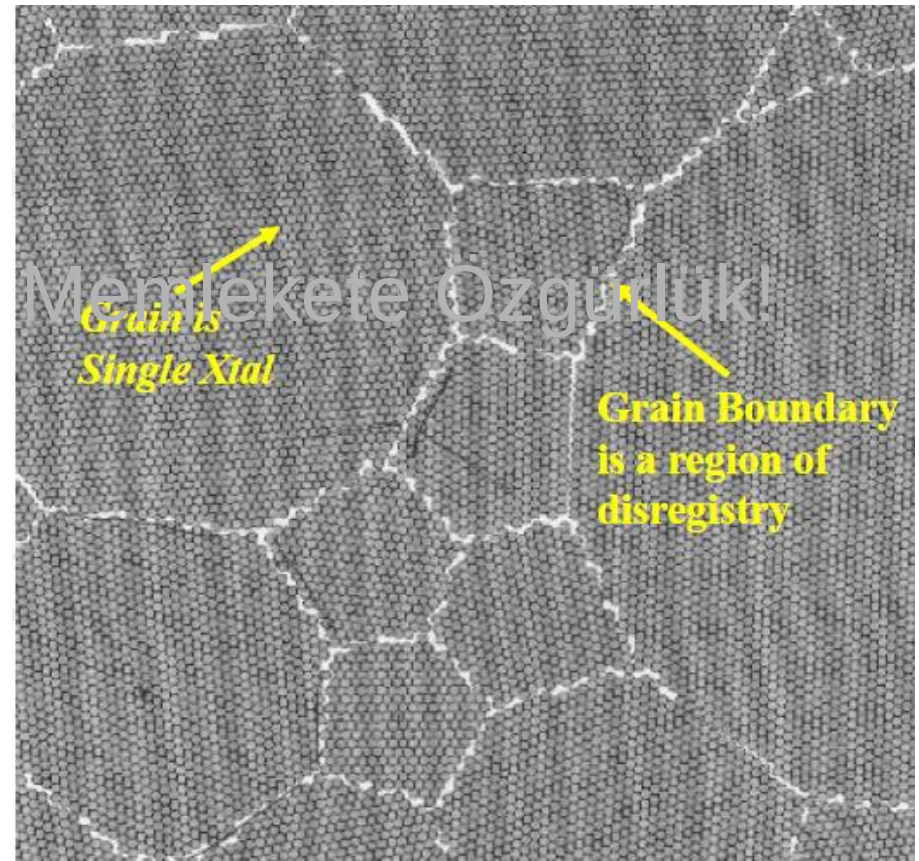
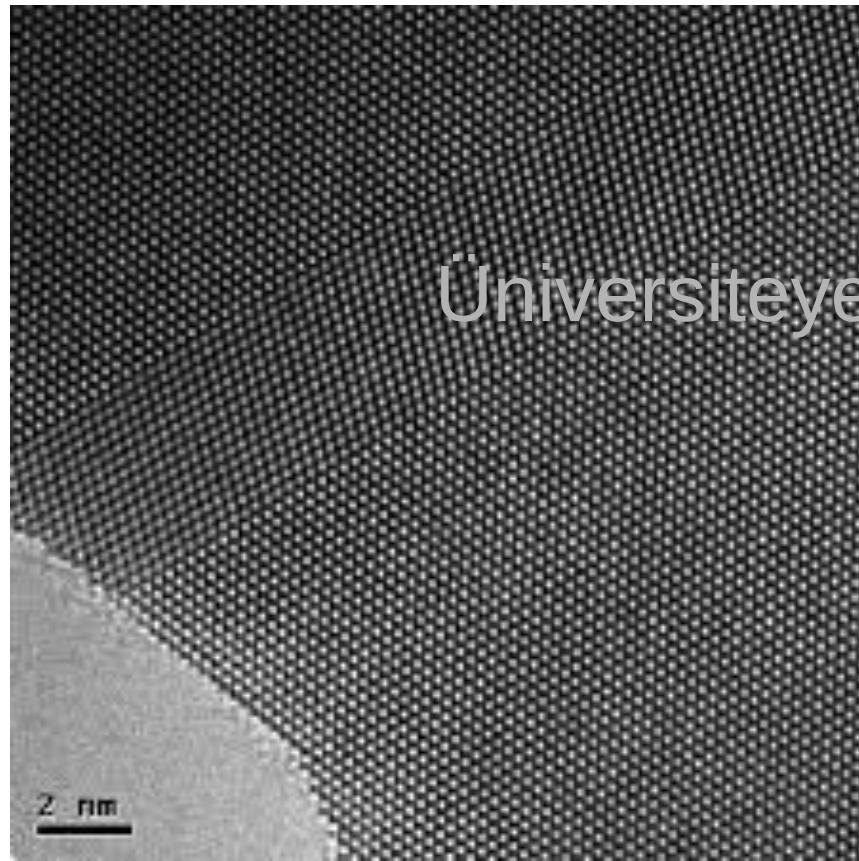
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Polycrystalline vs. Single Crystal



- Many single Xtals aggregated together = Polycrystalline material
 - Property of PolyXtal = Single Xtal Property + Grain Boundary property
- But...the properties are not necessarily additive in a straightforward manner!!!*

Atoms under e-microscopes



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Crystals as Building Blocks

- Some engineering applications require single crystals:
 - diamond single crystals for abrasives
 - turbine blades



(Courtesy Martin Deakins, GE Superabrasives, Worthington, OH. Used with permission.)

Fig. 8.33(c), *Callister & Rethwisch 8e.* (Fig. 8.33(c) courtesy of Pratt and Whitney).



- Properties of crystalline materials often related to crystal structure.
 - Ex: Quartz fractures more easily along some crystal planes than others.



(Courtesy P.M. Anderson)

Polycrystals

- Most engineering materials are polycrystals.



Anisotropic

Adapted from Fig. K, color inset pages of *Callister 5e*.
(Fig. K is courtesy of Paul E. Danielson, Teledyne Wah Chang Albany)

Isotropic

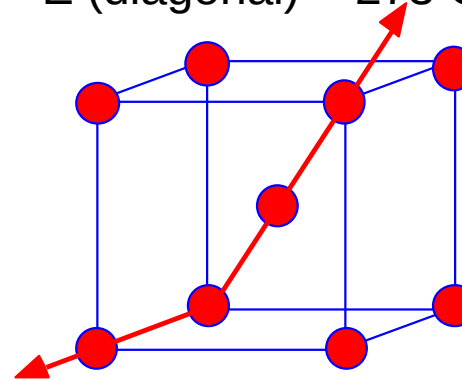
- Nb-Hf-W plate with an electron beam weld.
- Each "grain" is a single crystal.
- If grains are randomly oriented, overall component properties are not directional.
- Grain sizes typically range from 1 nm to 2 cm (i.e., from a few to millions of atomic layers).

Single vs Polycrystals

- Single Crystals

- Properties vary with direction: **anisotropic**.
- Example: the modulus of elasticity (E) in BCC iron:

E (diagonal) = 273 GPa

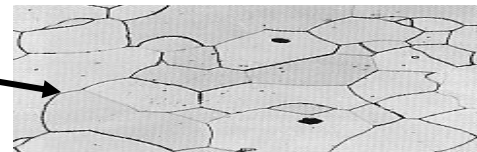
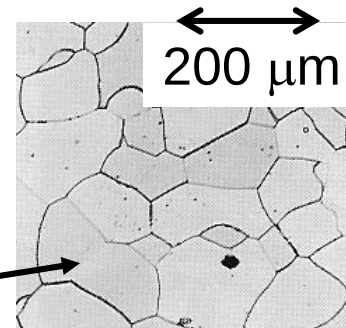


Data from Table 3.3, *Callister & Rethwisch 8e*. (Source of data is R.W. Hertzberg, *Deformation and Fracture Mechanics of Engineering Materials*, 3rd ed., John Wiley and Sons, 1989.)

- Polycrystals

- Properties may/may not vary with direction.
- If grains are randomly oriented: **isotropic**. ($E_{\text{poly iron}} = 210 \text{ GPa}$)
- If grains are **textured**, anisotropic.

E (edge) = 125 GPa



Adapted from Fig. 4.14(b), *Callister & Rethwisch 8e*. (Fig. 4.14(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC [now the National Institute of Standards and Technology, Gaithersburg, MD].)

Polymorphism

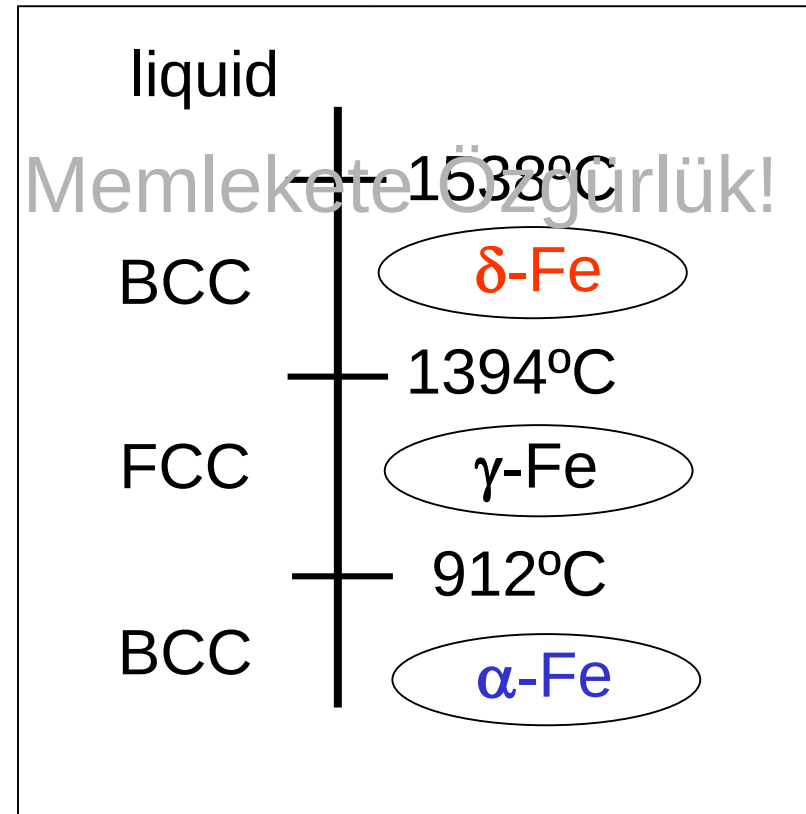
- Two or more distinct crystal structures for the same material (allotropy/polymorphism)

titanium
 α , β -Ti

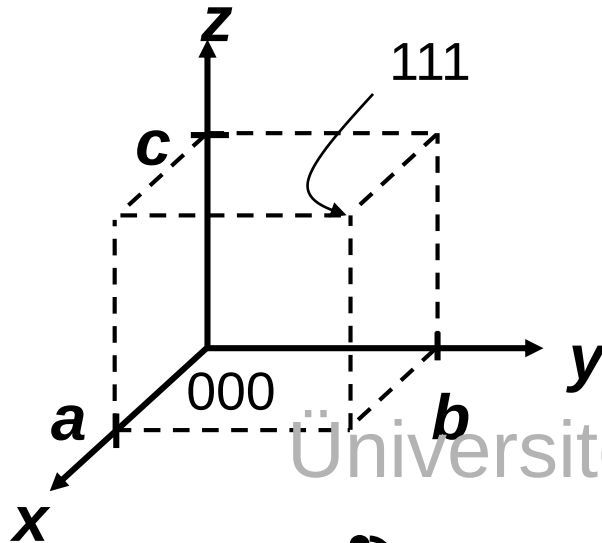
carbon
diamond, graphite

The metallurgy of **titanium** is dominated by the **crystallographic** transformation which takes place in the pure metal at 882° C. Below this temperature, pure **titanium** has a hexagonal close packed **structure** known as **alpha** (α); above it, the **structure** is body centred cubic and termed **beta** (β)

iron system



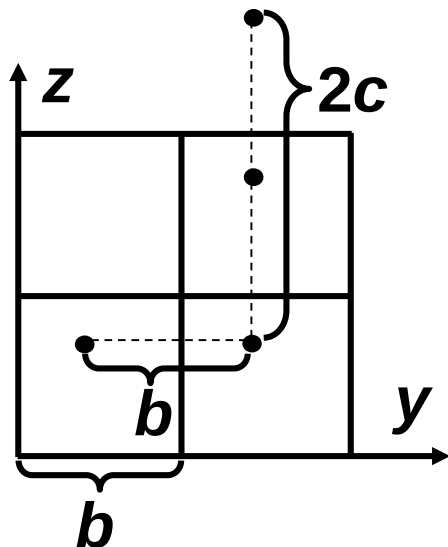
Point Coordinates



Point coordinates for unit cell center are

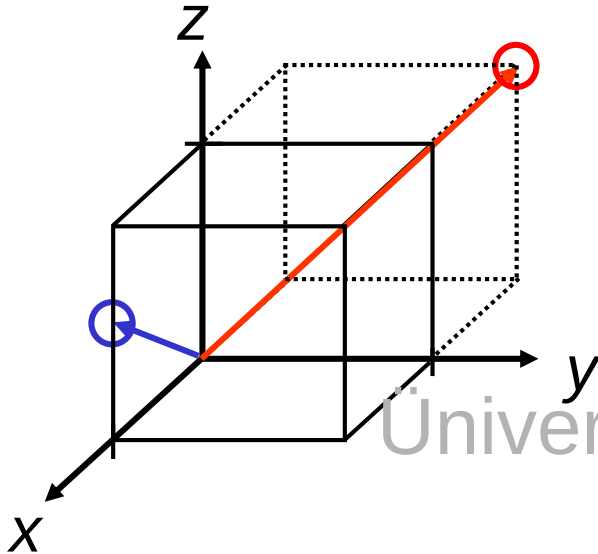
$$a/2, b/2, c/2 \quad \frac{1}{2} \frac{1}{2} \frac{1}{2}$$

Point coordinates for unit cell corner are 111



Translation: integer multiple of lattice constants $\rightarrow$ identical position in another unit cell

Crystallographic Directions



Algorithm

1. Vector repositioned (if necessary) to pass through origin.
2. Read off projections in terms of unit cell dimensions a , b , and c
3. Adjust to smallest integer values
4. Enclose in square brackets, no commas

$[uvw]$

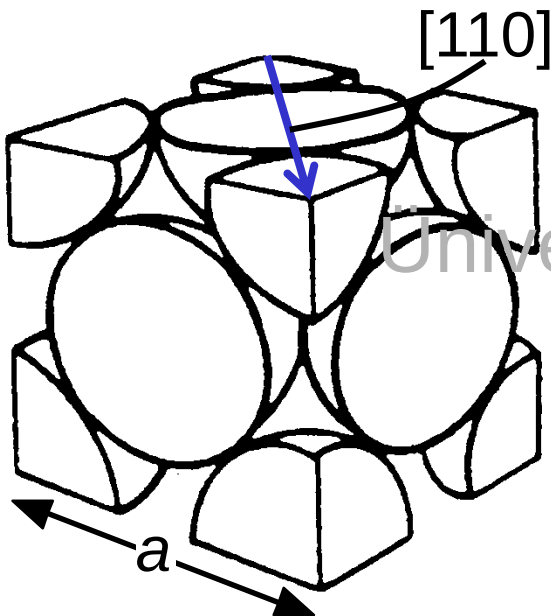
ex: $1, 0, \frac{1}{2} \Rightarrow 2, 0, 1 \Rightarrow [201]$

$-1, 1, 1 \Rightarrow [\bar{1}11]$ where overbar represents a negative index

families of directions $\langle uvw \rangle$

Linear Density

- Linear Density of Atoms $\equiv$ LD =
$$\frac{\text{Number of atoms}}{\text{Unit length of direction vector}}$$



Adapted from
Fig. 3.1(a),
Callister &
Rethwisch 8e.

ex: linear density of Al in [110]
direction

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 $a = 0.405 \text{ nm}$

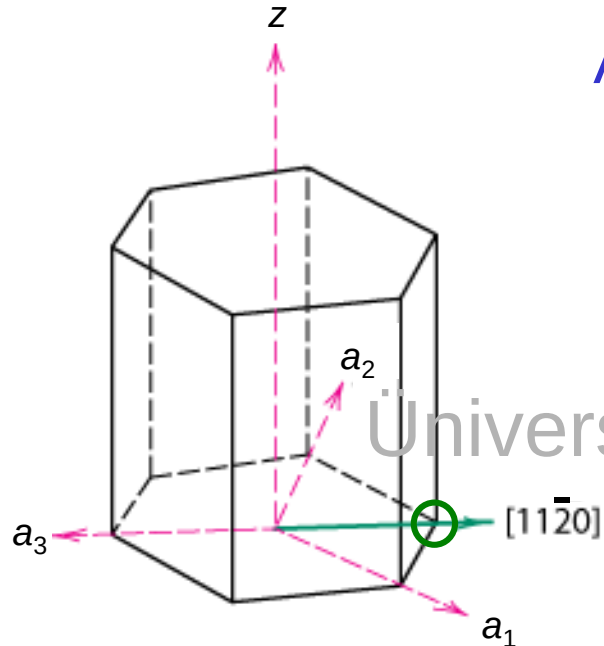
$$\text{LD} = \frac{\text{\# atoms}}{\text{length}} = \frac{2}{\sqrt{2}a} = 3.5 \text{ nm}^{-1}$$

HCP Crystallographic Directions

Algorithm

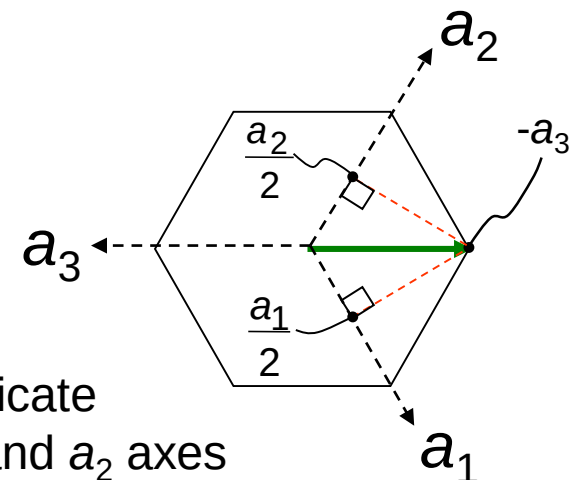
1. Vector repositioned (if necessary) to pass through origin.
2. Read off projections in terms of unit cell dimensions a_1 , a_2 , a_3 , or c
3. Adjust to smallest integer values
4. Enclose in square brackets, no commas

$[uvw]$



Adapted from Fig. 3.8(a),
Callister & Rethwisch 8e.

ex: $\frac{1}{2}, \frac{1}{2}, -1, 0 \Rightarrow [11\bar{2}0]$



dashed red lines indicate
projections onto a_1 and a_2 axes

HCP Crystallographic Directions

- Hexagonal Crystals

- 4 parameter Miller-Bravais lattice coordinates are related to the direction indices (i.e., $u'v'w'$) as follows.

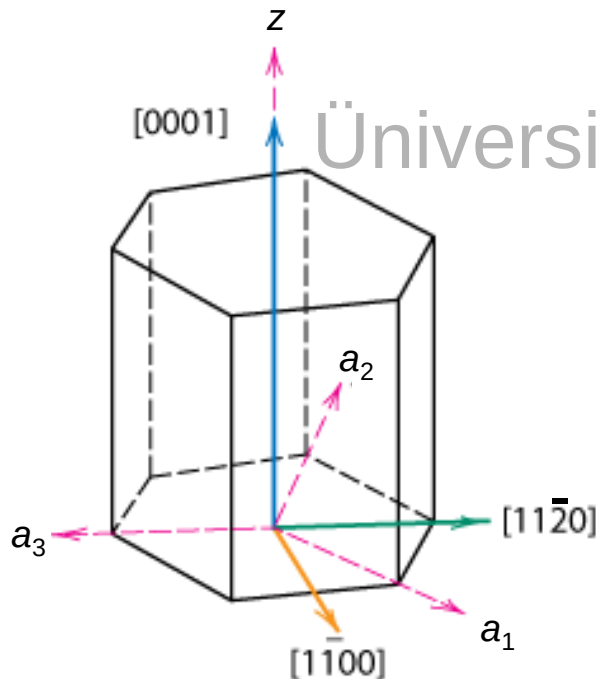


Fig. 3.8(a), Callister & Rethwisch 8e.

$$[u'v'w'] \rightarrow [uvw]$$

$$u = \frac{1}{3}(2u' - v')$$

$$v = \frac{1}{3}(2v' - u')$$

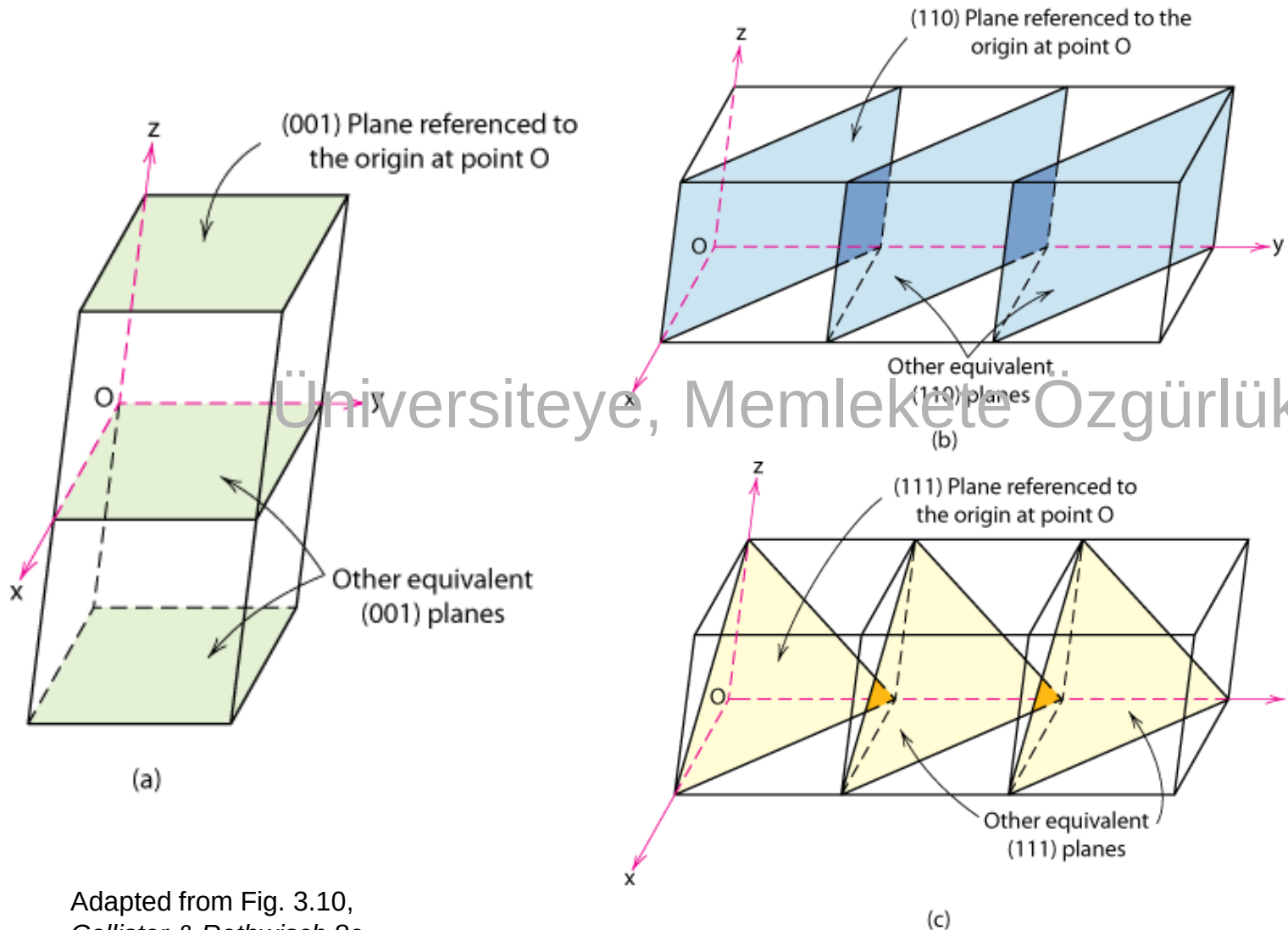
$$t = -(u + v)$$

$$w = w'$$

Crystallographic Planes

- Miller Indices: Reciprocals of the (three) axial intercepts for a plane, cleared of fractions & common multiples. All parallel planes have same Miller indices.
- Algorithm
 1. Read off intercepts of plane with axes in terms of a , b , c
 2. Take reciprocals of intercepts
 3. Reduce to smallest integer values
 4. Enclose in parentheses, no commas i.e., (hkl)

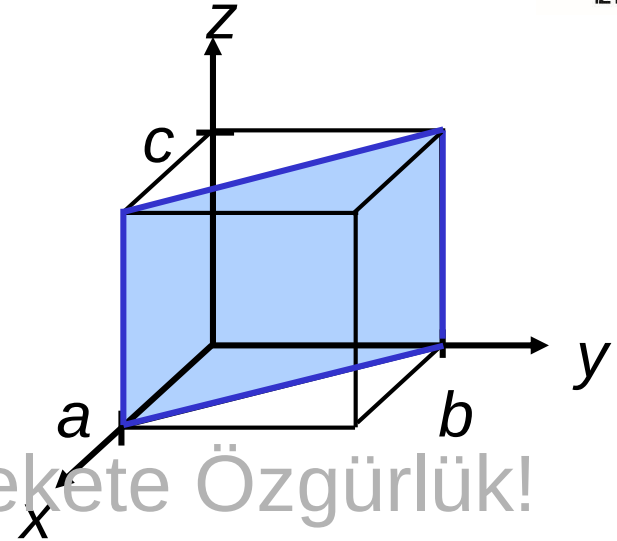
Crystallographic Planes



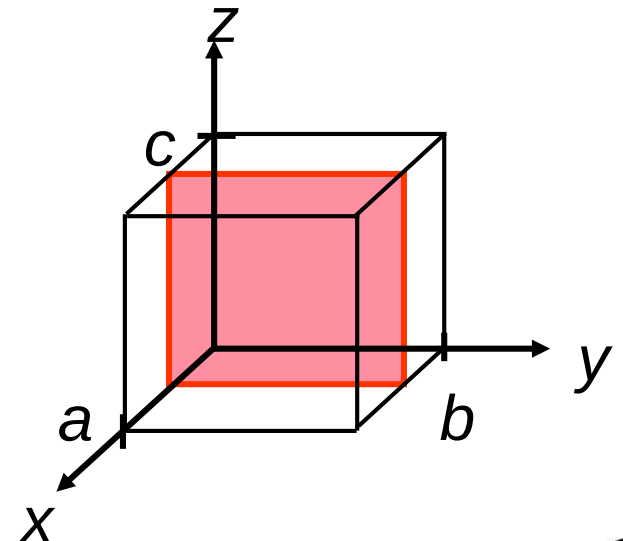
Adapted from Fig. 3.10,
Callister & Rethwisch 8e.

Crystallographic Planes

<u>example</u>	<i>a</i>	<i>b</i>	<i>c</i>
1. Intercepts	1	1	∞
2. Reciprocals	1/1	1/1	1/ ∞
	1	1	0
3. Reduction	1	1	0
4. Miller Indices	(110)		

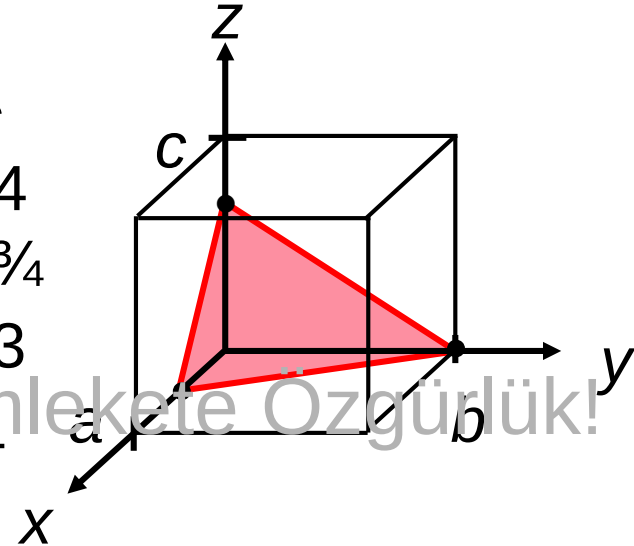


<u>example</u>	<i>a</i>	<i>b</i>	<i>c</i>
1. Intercepts	1/2	∞	∞
2. Reciprocals	1/1/2	1/ ∞	1/ ∞
	2	0	0
3. Reduction	2	0	0
4. Miller Indices	(100)		



Crystallographic Planes

<u>example</u>	<i>a</i>	<i>b</i>	<i>c</i>
1. Intercepts	1/2	1	3/4
2. Reciprocals	1/1/2	1/1	1/3/4
	2	1	4/3
3. Reduction	6	3	4
4. Miller Indices	(634)		

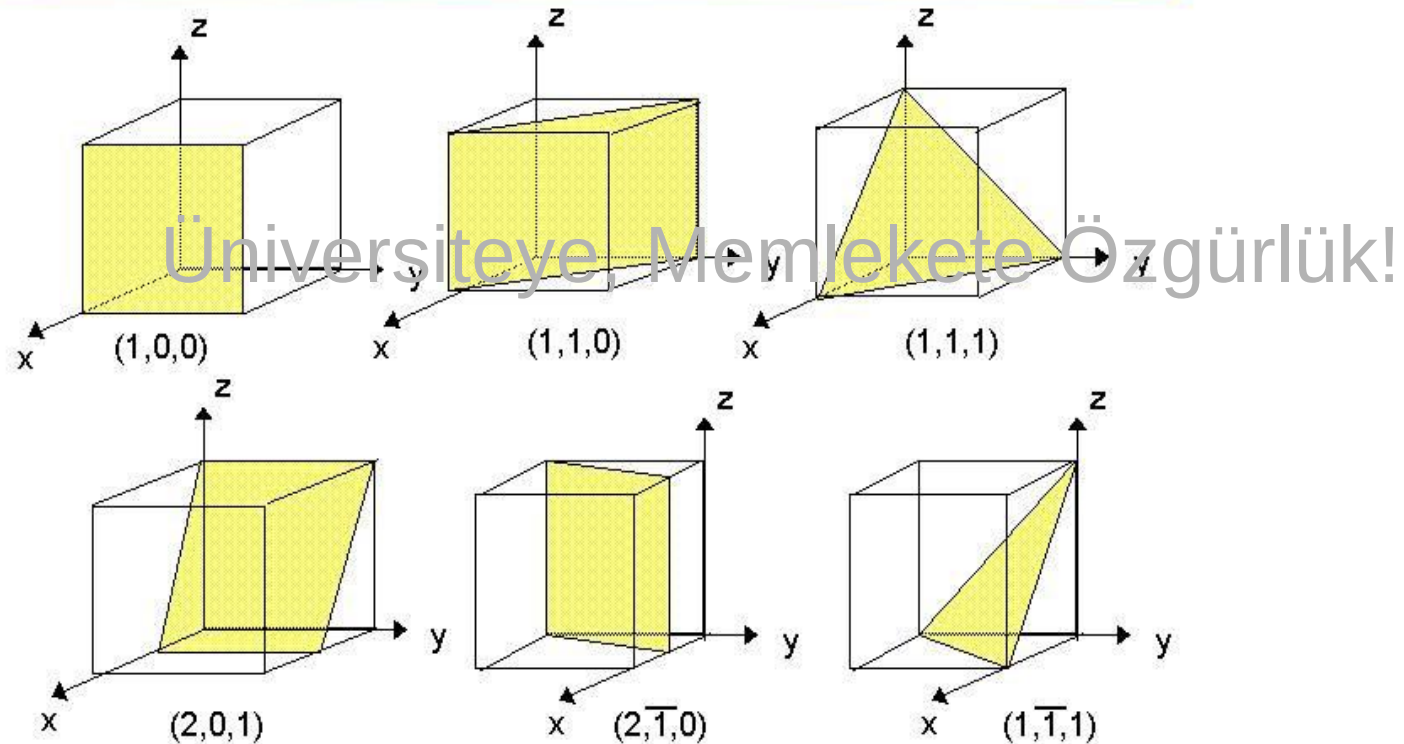


Family of Planes $\{hkl\}$

Ex: $\{100\} = (100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1})$

Examples of cubic symmetry

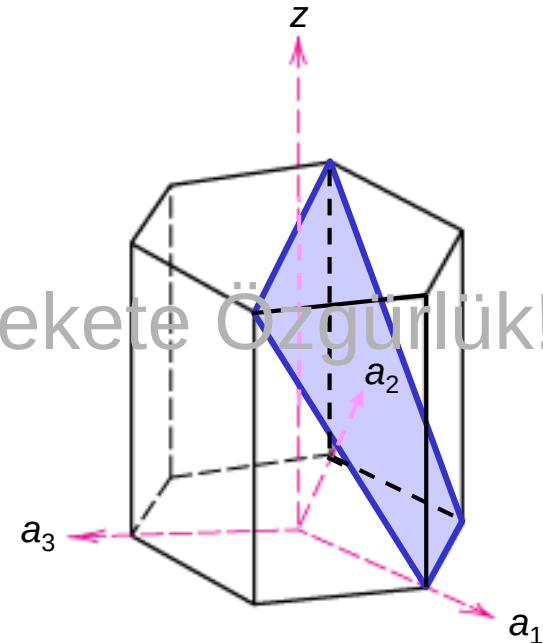
Some examples



Crystallographic Planes (HCP)

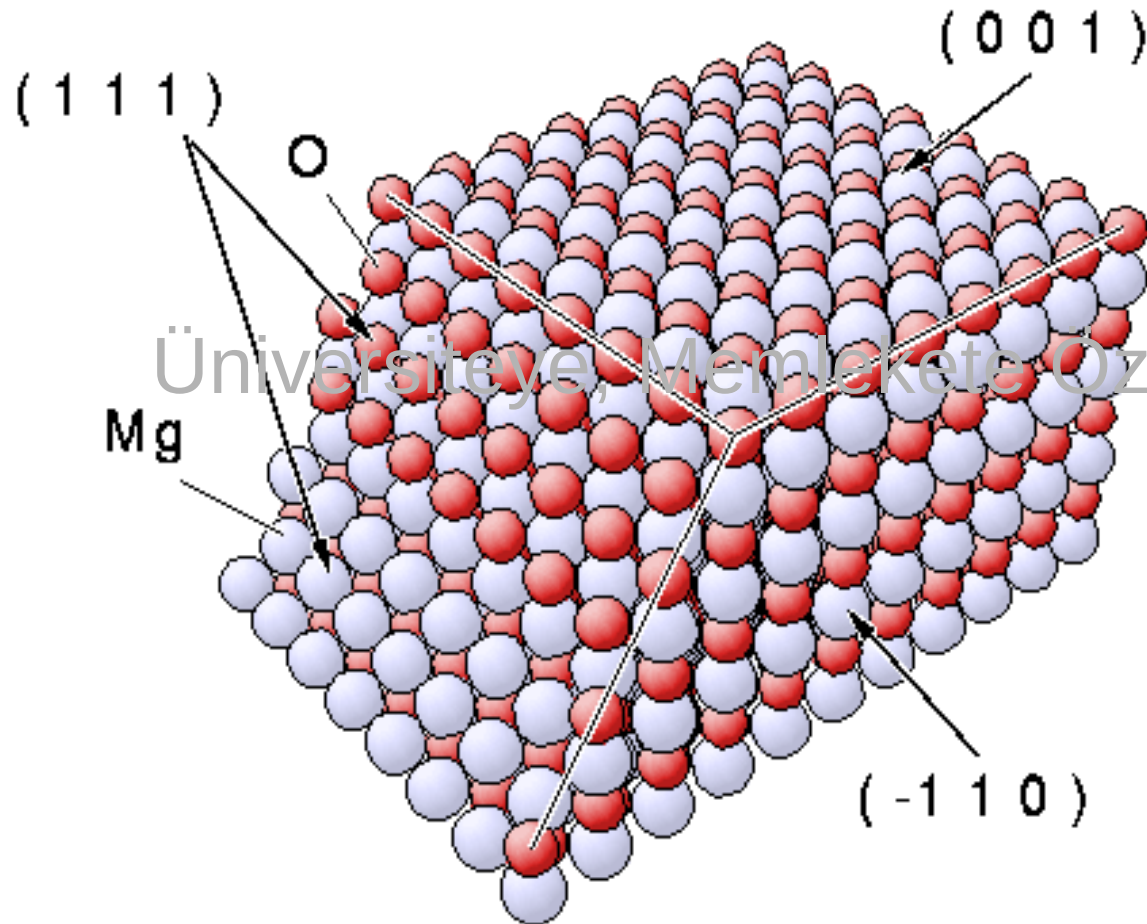
- In hexagonal unit cells the same idea is used

<u>example</u>	a_1	a_2	a_3	c
1. Intercepts	1	∞	-1	1
2. Reciprocals	1	$1/\infty$	-1	1
	1	0	-1	1
3. Reduction	1	0	-1	1
4. Miller-Bravais Indices	$(10\bar{1}1)$			

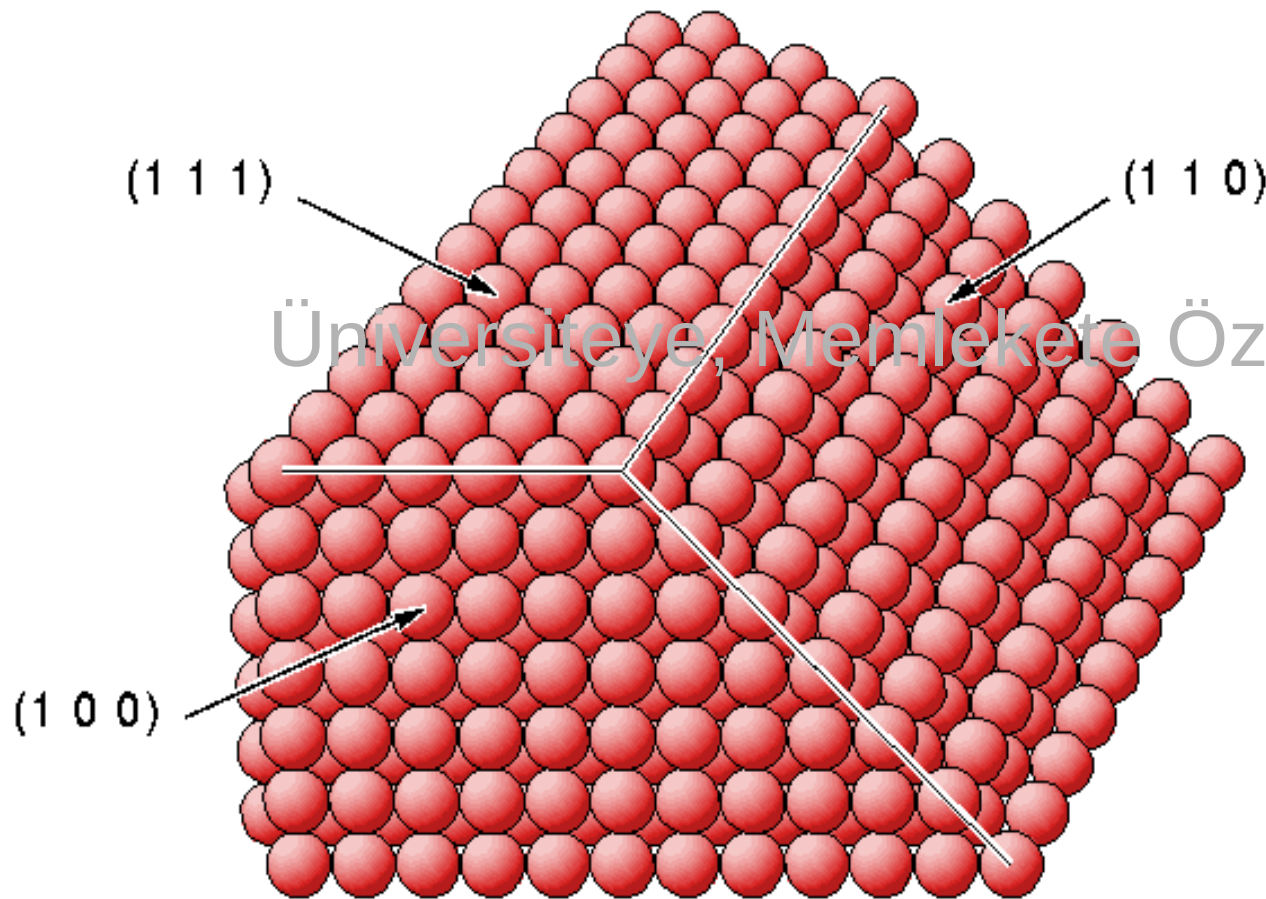


Adapted from Fig. 3.8(b),
Callister & Rethwisch 8e.

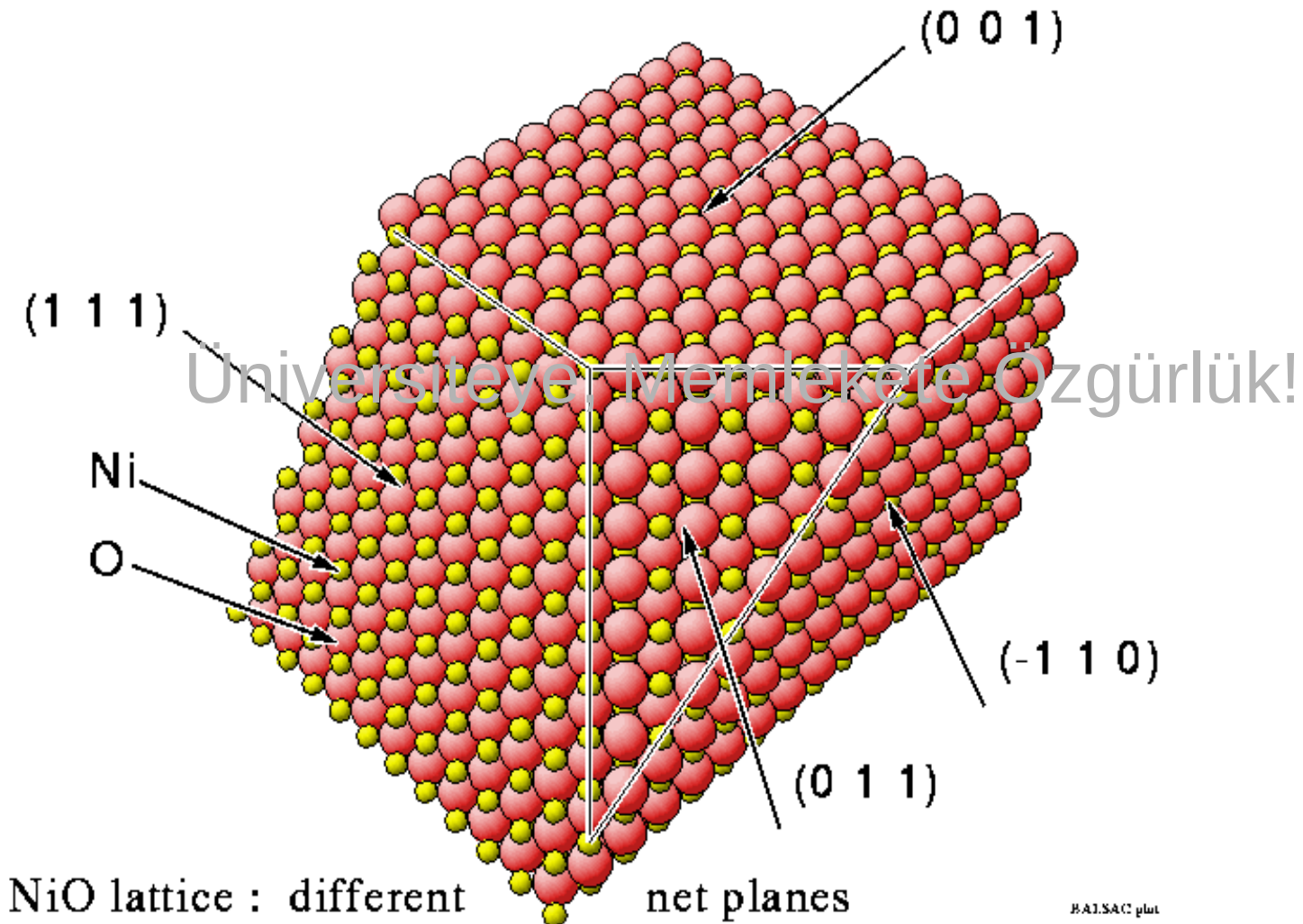
MgO (Rocksalt) crystal



FCC lattice: different planes



NiO (Rocksalt) structure



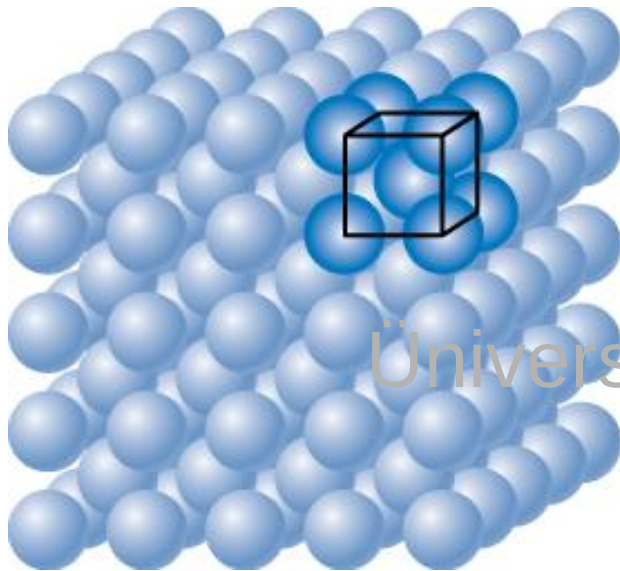
BA1SAC plot

Crystallographic Planes

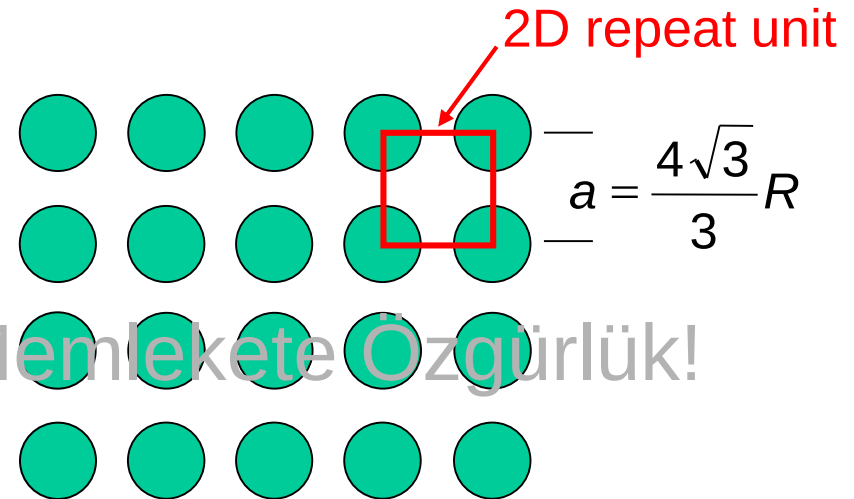
- We want to examine the atomic packing of crystallographic planes
- Iron foil can be used as a catalyst. The atomic packing of the exposed planes is important.
a) Draw (100) and (111) crystallographic planes for Fe.
b) Calculate the planar density for each of these planes.

Planar Density of (100) Iron

Solution: At $T < 912^{\circ}\text{C}$ iron has the BCC structure.



(100)



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Adapted from Fig. 3.2(c), Callister & Rethwisch 8e.

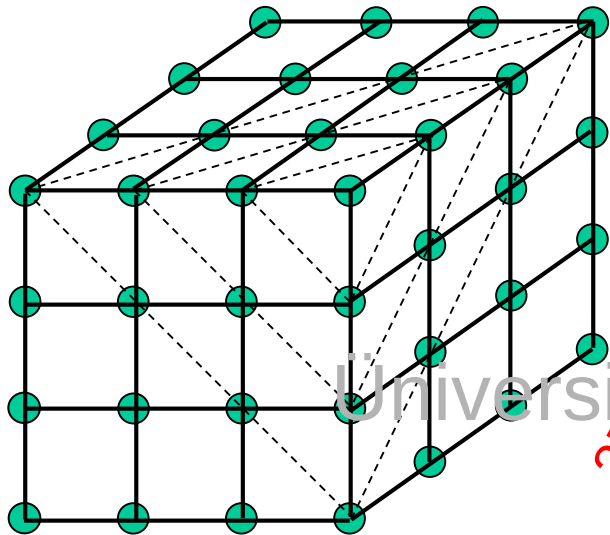
Radius of iron $R = 0.1241 \text{ nm}$

$$\text{Planar Density} = \frac{\frac{\text{atoms}}{\text{2D repeat unit}}}{\frac{\text{area}}{\text{2D repeat unit}}} = \frac{1}{\left(\frac{4\sqrt{3}}{3}R\right)^2} = 12.1 \frac{\text{atoms}}{\text{nm}^2} = 1.2 \times 10^{19} \frac{\text{atoms}}{\text{m}^2}$$

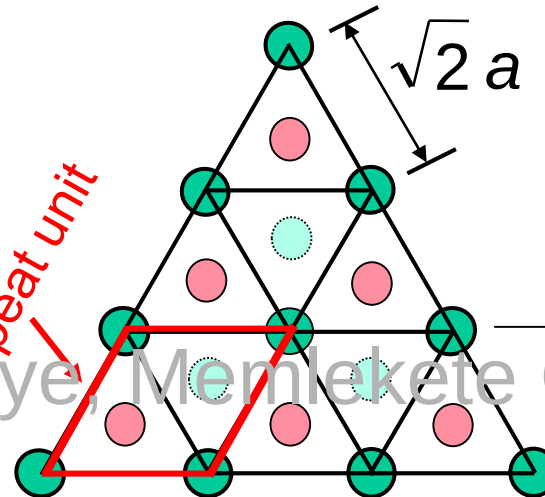
Planar Density of (111) Iron

Solution (cont): (111) plane

1 atom in plane/ unit surface cell



2D repeat unit



- atoms in plane
- atoms above plane
- atoms below plane

$$\text{area} = \sqrt{2} ah = \sqrt{3} a^2 = \sqrt{3} \left(\frac{4\sqrt{3}}{3} R \right)^2 = \frac{16\sqrt{3}}{3} R^2$$

atoms
2D repeat unit

1

Planar Density =

area
2D repeat unit

$$\frac{1}{\frac{16\sqrt{3}}{3} R^2}$$

$$= 7.0 \frac{\text{atoms}}{\text{nm}^2} =$$

$$0.70 \times 10^{19} \frac{\text{atoms}}{\text{m}^2}$$

SUMMARY

- Atoms may assemble into **crystalline** or **amorphous** structures.
- Common metallic crystal structures are **FCC**, **BCC**, and **HCP**. **Coordination number** and **atomic packing factor** are the same for both FCC and HCP crystal structures.
- We can predict the **density** of a material, provided we know the **atomic weight**, **atomic radius**, and **crystal geometry** (e.g., FCC, BCC, HCP).
- **Crystallographic points**, **directions** and **planes** are specified in terms of indexing schemes.
Crystallographic directions and planes are related to **atomic linear densities** and **planar densities**.

SUMMARY

- Materials can be **single crystals** or **polycrystalline**.
Material properties generally vary with single crystal orientation (i.e., they are **anisotropic**), but are generally non-directional (i.e., they are **isotropic**) in polycrystals with randomly oriented grains.
- Some materials can have more than one crystal structure. This is referred to as **polymorphism** (or **allotropy**).